DHRUV DUBEY

AI Engineer | Agentic AI, LLM Systems & RAG | College Park, MD
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EDUCATION

University of Maryland, College Park — M.S., Data Science (GPA 4.00/4.00) *Aug 2025 – May 2027*

MIT-ADT University — B.Tech, Computer Science & Engineering (GPA 3.33/4.00) *2021 – 2025*

TECHNICAL SKILLS

Languages: Python, SQL, C++, R
Agents & LLMs: Agentic AI, Multi-Agent Systems, LangGraph, LangChain, CrewAI, Model Context Protocol (MCP), Tool Calling, Prompt & Context Engineering, RAG, Vector Databases, Embeddings, LoRA/QLoRA Fine-Tuning, Open-Weight LLMs (Qwen3 via Ollama), Snowflake Cortex
Evaluation: LLM & Agent Evaluation, Golden Datasets, CI Eval Gates, Crash-Recovery Benchmarks, Regression Testing
Backend & MLOps: FastAPI, REST APIs, Pydantic, Microservices, Docker, Kubernetes (HPA, Probes), GitHub Actions CI/CD, MLflow, AWS, SLURM/HPC
ML: PyTorch, TensorFlow, Scikit-learn, Deep Learning, NLP, Computer Vision, Time Series, Anomaly Detection
Data: Snowflake, PostgreSQL, SQLite, Redis, MongoDB, Neo4j, Supabase, Power BI, Pandas, NumPy

EXPERIENCE

Allegis Global Solutions — AI Automation Engineer Intern *Jun 2026 – Sep 2026*
- Built a multi-agent triage system on Snowflake Cortex (Agents, Search, Analyst) that traces pipeline failures through a dependency graph to the broken stored procedure; root-cause diagnosis ~65% faster (weeks to days).
- Engineered semantic views and a YAML instruction layer from prior tickets so the agent handles new failure patterns without retraining.
- Deployed it into analysts' Power BI workflow as a chat interface, cutting fix-proposal time from hours to minutes.

University of Maryland — Research Intern, Neuro-Symbolic AI *Sep 2025 – Sep 2026*
- Co-developed a six-stage reproducibility audit of neuro-symbolic AI: screened 5,497 records to 1,304 studies and reproduced 85 codebases on the Zaratan HPC cluster (SLURM, Docker).
- Found inconsistencies in setups, hyperparameters and evaluation methods across reasoning frameworks; co-authored the paper, now under review at IEEE TNNLS.

Fortytwo Labs — Machine Learning Intern *Feb 2025 – Jul 2025*
- Built TensorFlow autoencoders on login behavior (+40% anomaly detection accuracy) and KMeans/DBSCAN clustering that surfaced 35% more outliers.
- Integrated the models with the Drools rule engine for automated risk scoring, cutting manual security reviews 33%.

Intrainz — Data Science Intern *May 2023 – Jul 2023*
- Built a crop recommendation system on 5,000+ records, raising decision-support precision 15% across 150 pilot farmers.

PROJECTS

Pipeline Incident Triage Graph GitHub
LangGraph, Qwen3 8B (Ollama), Model Context Protocol, FastAPI, Pydantic, Docker, Kubernetes, GitHub Actions
- Open-source LangGraph triage: an LLM investigator with read-only tools walks the pipeline DAG to the broken procedure; a rule-based tracer cross-checks it, and on 20 held-out incidents stopped all 7 wrong Qwen3 8B diagnoses before approval.
- CI gated on two 100-incident eval sets: root-cause accuracy 1.00; noisy logs drop category accuracy to 0.81, which the gate tracks. Ships an MCP server and Kubernetes manifests.

Keel: Approval-Bound Agent Runtime GitHub
Python, Qwen3 8B (Ollama), FastAPI, SQLite, Model Context Protocol, Pydantic, Docker, Kubernetes, GitHub Actions
- Runtime that holds an LLM fixer to exactly the human-approved plan (nothing unapproved, twice or skipped) and logs every call so a crashed run resumes; runs the triage investigator as a read-only first agent: 28 of 40 root causes on Qwen3 8B.
- 200-run crash benchmark: 0 duplicated side effects vs 145 on restart; with the triage agent inside it, 75 fixes killed mid-apply gave 0 duplicated vs 61.
- Added a loop that tunes it against its own benchmark: human interruptions after a crash 30 of 75 to 0, nothing duplicated or dropped, 4 of 10 candidates rejected by a safety gate.

Krishi Mitra — Multilingual Agricultural RAG Assistant GitHub
- Fine-tuned LLaMA 3.2 3B with RAG answering farmers in Hindi, Marathi and English; +20% query-resolution accuracy across a 300-farmer pilot.

Smart Plant Health Monitor GitHub
- YOLOv5, ResNet and LSTM system classifying plant health at 90% accuracy, served as a Dockerized FastAPI microservice.

PUBLICATIONS

- Neuro-Symbolic AI Reproducibility Review — University of Maryland; arXiv:2608.26236; under review, IEEE TNNLS.
- Krishi Mitra: LLM-Driven QA System for Indian Farmers — MIT-ADT University.